

Before doing assignment, students should create two threads in C where each thread prints some message with thread id.

Assignment-01:

- a) In this part student should demonstrate race condition for 2 threads using programming in C.
- b) In this part student should Fix the Race Condition with Mutex shown in above for 2 threads using programming in C.
- c) Repeat the above for more than 2 threads (say for 5 threads) using programming in C.

Assignment-02:

- a) Implement solution for producer-consumer problem using only one mutex using programming in C. Analyze what are the problems?
- b) Implement solution for producer-consumer problem using one mutex and conditional wait without semaphore using programming in C. Analyze your solution.
- c) Implement solution for producer-consumer problem using semaphores using programming in C. Analyze your solution.